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Level Up

MACHINE LEARNING

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Computer SCIENCE

Regression Performance Metrics

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Ref: Medium, Web

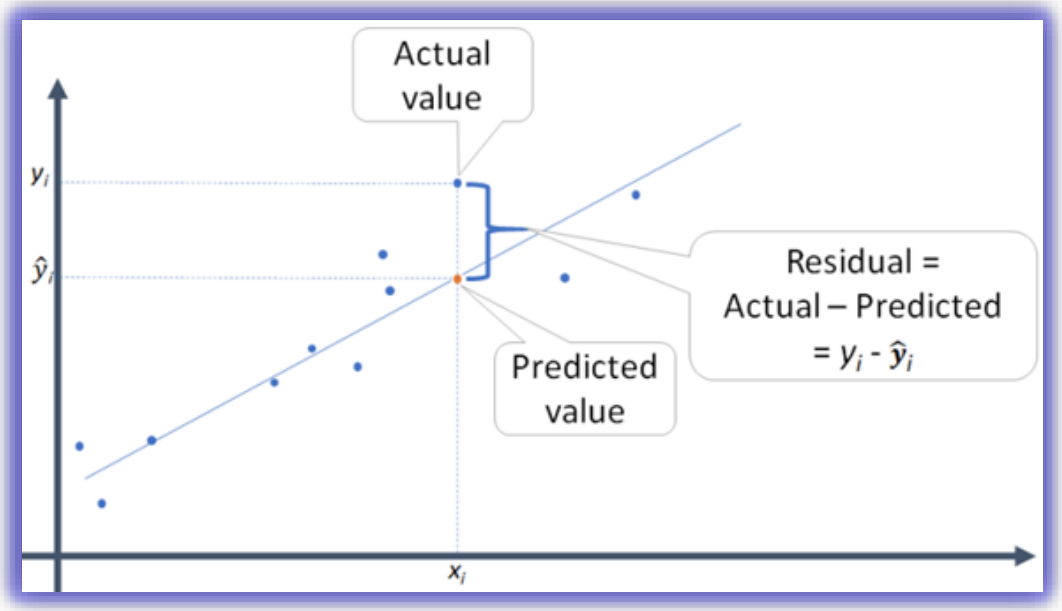
Residuals

Residuals are the differences **between** observed and predicted values generated by a statistical model.

Residual = Actual – Predicted = y_i - $\hat{y}_i$

In regression analysis, the residual provides insight into the model's **accuracy** and its underlying assumptions.

If residuals are **small**, it implies that the model that produced them does a **good** job at predicting our output of interest. Conversely, if these residuals are generally **large**, it implies that model is a **poor** estimator.



Regression Model

Regression metrics are **quantitative** measures used to evaluate the nice of a regression model. **Evaluation metrics** are a measure of how well a model performs and how well it approximates the relationship.

In the context of **regression**, ***Models*** refer to mathematical **Equations** used to describe the relationship between two variables. In general, these models deal with the ***prediction*** and ***estimation*** of values of interest in our data called **outputs**. Models will look at other aspects of the data, called **inputs,** that we believe affect the outputs and use them to generate estimated outputs.

These inputs and outputs have many names. Inputs can also be called **independent** variables or **predictors**, while outputs are also known as **responses** or **dependent** variables. Simply speaking, models are just functions where the outputs are some ***function*** of the inputs. The **linear** part of linear regression refers to the fact that a linear regression model is described mathematically in the form:

Linear Regression: Single Variable

$$\underbrace{\boxed{\hat{y}}}_{\text{Predicted output}} = \underbrace{\beta_0 + \beta_1}_{\text{Coefficients}} \underbrace{\boxed{x}}_{\text{Input}} + \underbrace{\boxed{\epsilon}}_{\text{Error}}$$

Linear Regression: Multiple Variables

$$\boxed{\hat{y}} = \beta_0 + \beta_1 \boxed{x_1} + \ldots + \beta_p \boxed{x_p} + \boxed{\epsilon}$$

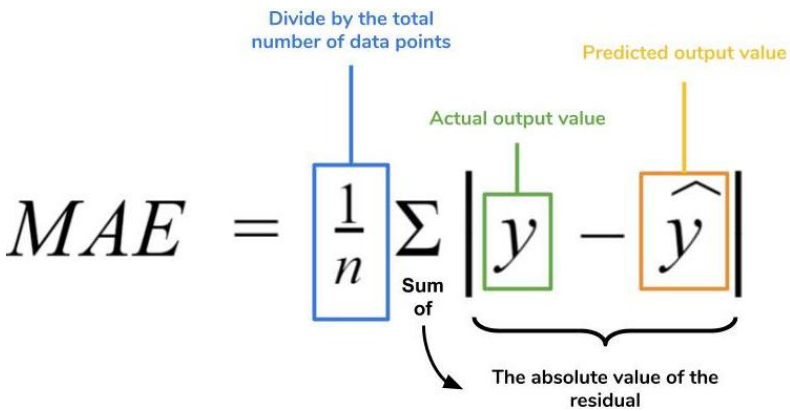
Types

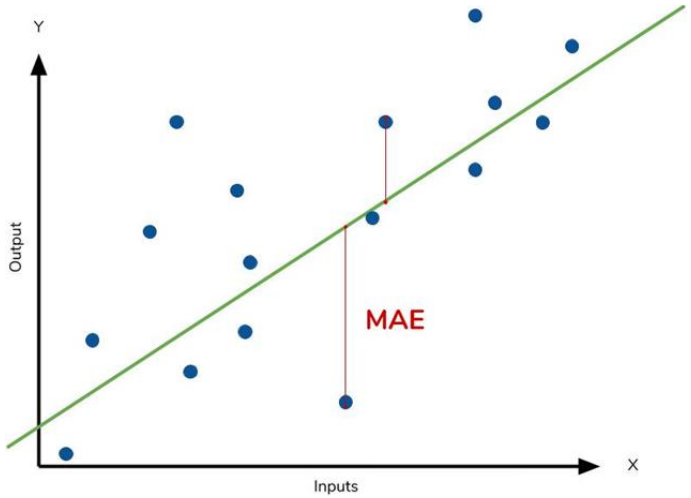
Some common regression metrics are:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- Mean Percentage Error (MPE)
- R-squared (R^2) Score
- Adjusted R-squared (R^2) Score

Mean Absolute Error (MAE)

MAE is simply the average of the absolute difference between the target value and the value predicted by the model.
Not preferred in cases where **outliers** are prominent.
MAE **does not penalize** large errors.



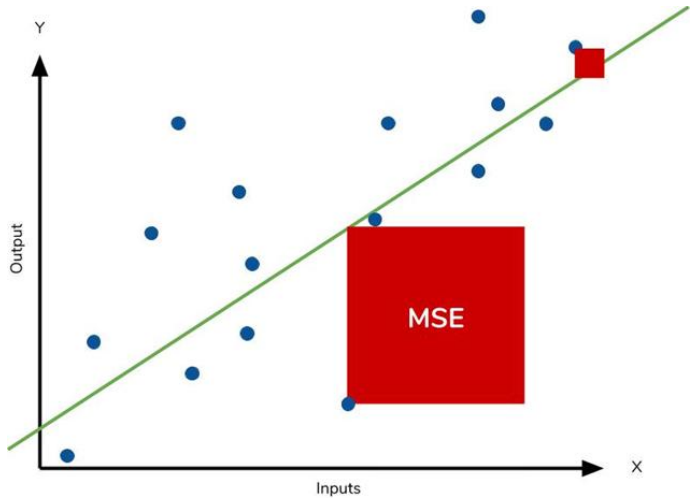


Mean Squared Error (MSE)

MSE has a **convex** shape. It is the average of the squared difference between the predicted and actual values.
Since it is differentiable and has a convex shape, it is easier to **optimize**.
MSE **penalizes** large errors.

$$MSE = \frac{1}{n}\Sigma\underbrace{\left(y - \hat{y}\right)^2}_{\text{The square of the difference between actual and predicted}}$$

$$RMSE = \sqrt{\frac{1}{n}\Sigma\left(y - \hat{y}\right)^2}$$



Mean Absolute Percentage Error (MAPE)

MAPE is the percentage equivalent of MAE.
The equation looks just like that of MAE, but with adjustments to convert everything into **percentages**.

Just as MAE is the average magnitude of error produced by your model, the MAPE is how far the model's predictions are off from their corresponding outputs on average. Like MAE, MAPE also has a clear interpretation since percentages are easier for people to conceptualize. Both MAPE and MAE are robust to the effects of outliers thanks to the use of absolute value.

Multiplying by 100% converts to percentage

$$MAPE = \frac{100\%}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{y} \right|$$

Each residual is scaled against the actual value

$\hat{y}$ is smaller than the actual value

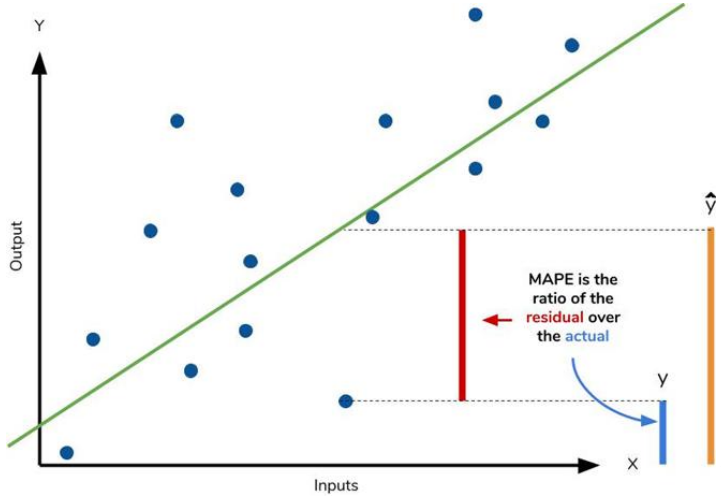
n = 1 $\hat{y}$ = 10 y = 20

MAPE = 50%

$\hat{y}$ is greater than the actual value

n = 1 $\hat{y}$ = 20 y = 10

MAPE = 100%

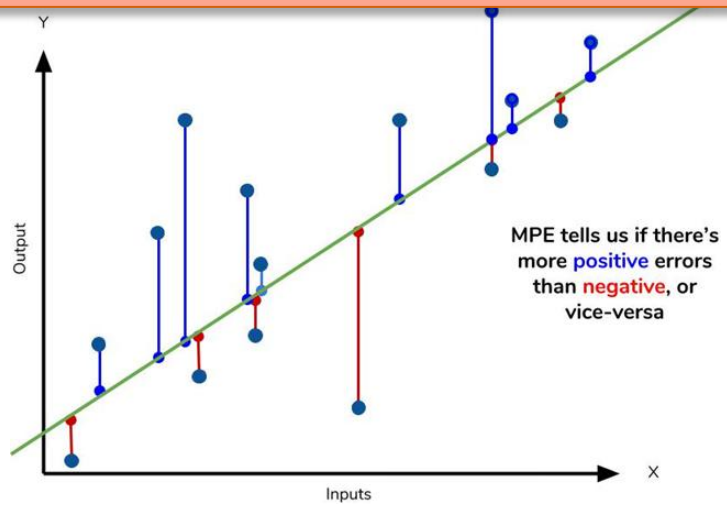


Mean Percentage Error (MPE)

The MPE equation is exactly like that of MAPE.
The only difference is that it **lacks** the **absolute** value operation.

Although the MPE lacks the absolute value operation, it is actually its absence that makes the MPE **useful**. Since **positive** and **negative** errors cancel each other out, we cannot make any statements about how well the model's predictions perform overall. However, if there are more -ve or +ve errors, this bias will show up in the MPE. Unlike MAE and MAPE, MPE is useful to us because it allows us to see if our model systematically **underestimates** (more **-ve** error) or **overestimates** (more **+ve** error).

$$MPE = \frac{100\%}{n} \Sigma \left(\frac{y - \hat{y}}{y} \right)$$



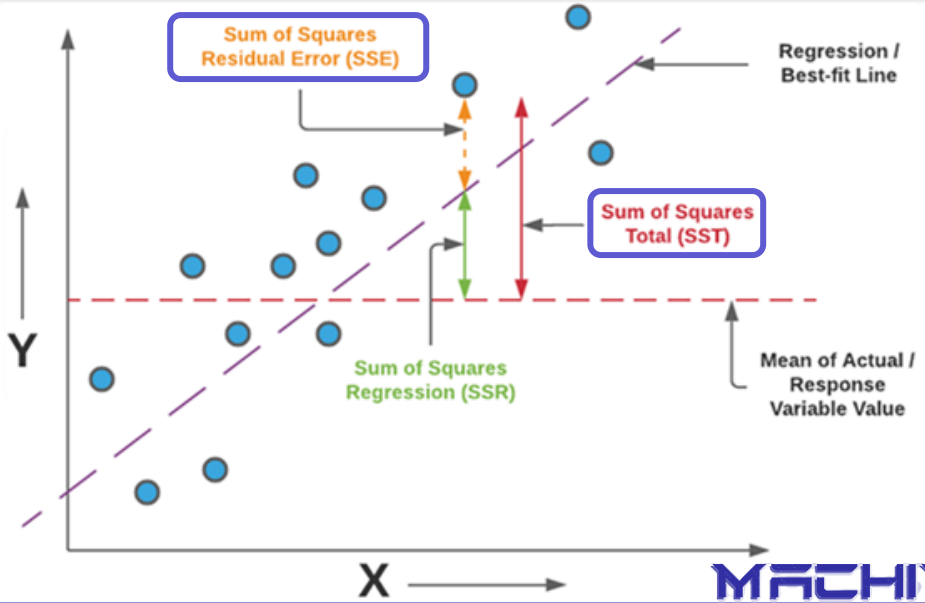
R-Squared Score

R^2 metric represents the part of the **variance** of the dependent variable explained by the independent variables of the model. It measures the **strength** of the relationship between your model and the dependent variable.

If the data points are very **close** to the regression line, then the model accounts for a **good** amount of variance, thus resulting in a **high** R^2 value. However, do not let the R^2 value fool you. A good model can have low R^2 value, and a biased model can have a high R^2 value as well. That is the reason you should make use of **residual** plots.

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

R^2 values start from 0 and Max up to 1. The **higher** the value of R^2, the closer to 1 the **better** the model.



Adjusted R-Squared Score

One **downside** of the R^2 score is that as **new features** are added to the data, the R^2 score starts to **increase** or stays **constant** but never **decreases** because it assumes that the variance of the data increases with more data.
But the problem is that when we add **extraneous features** to the dataset, R^2 sometimes starts to increase. This is **incorrect**.

Therefore, **Adjusted** R^2 emerged as R^2's alternative to control this situation. The mathematical formula for computing adjusted R^2 is given below:

$$Adjusted\ R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

- Unlike R-squared, Adjusted R-squared can **decrease** with the **addition** of **less significant** variables, thus resulting in a more **reliable** and **accurate** evaluation.
- Adjusted R-squared is an efficient evaluation metric and can **correlate** the variables **more accurately** than R-squared.